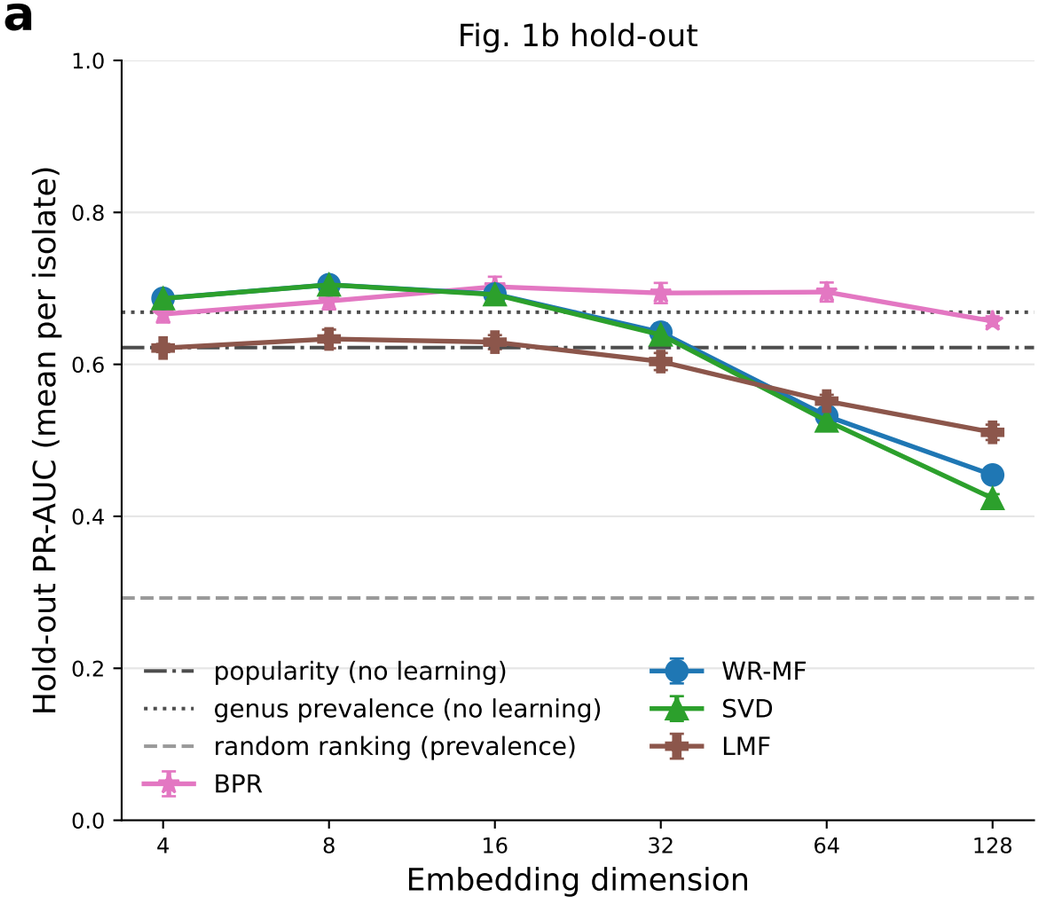
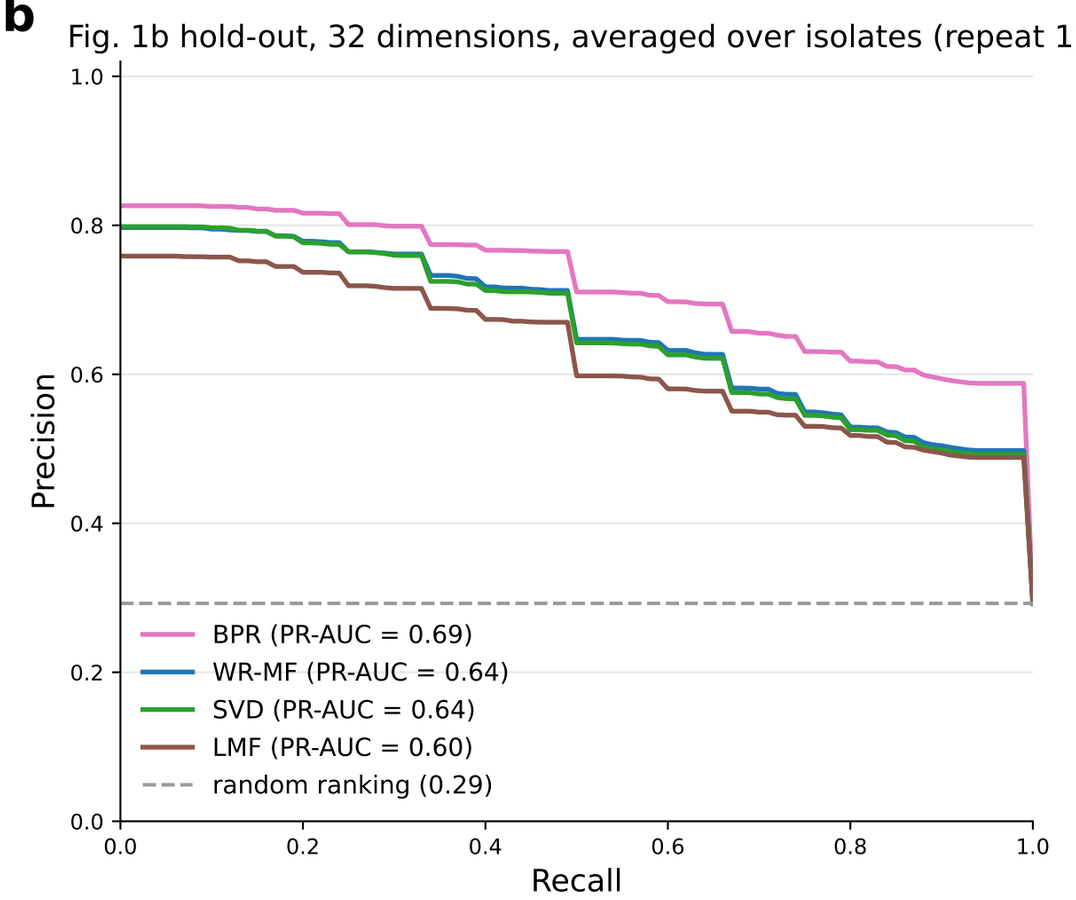
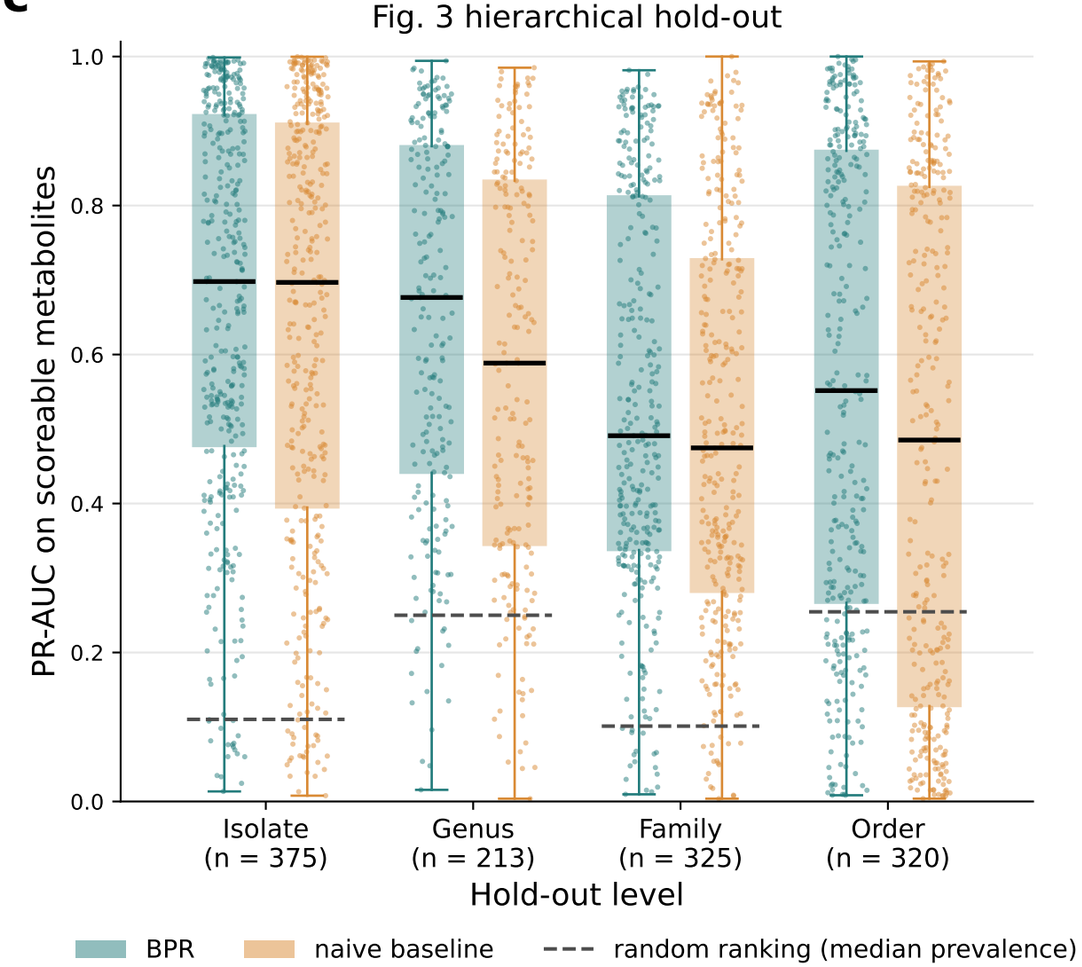
a


Fig. 1b hold-out
Hold-out PR-AUC (mean per isolate)
1.0
0.8
0.6
0.4
0.2
0.0
4
8
16
32
64
128
Embedding dimension
popularity (no learning)
genus prevalence (no learning)
random ranking (prevalence)
BPR
WR-MF
SVD
LMF

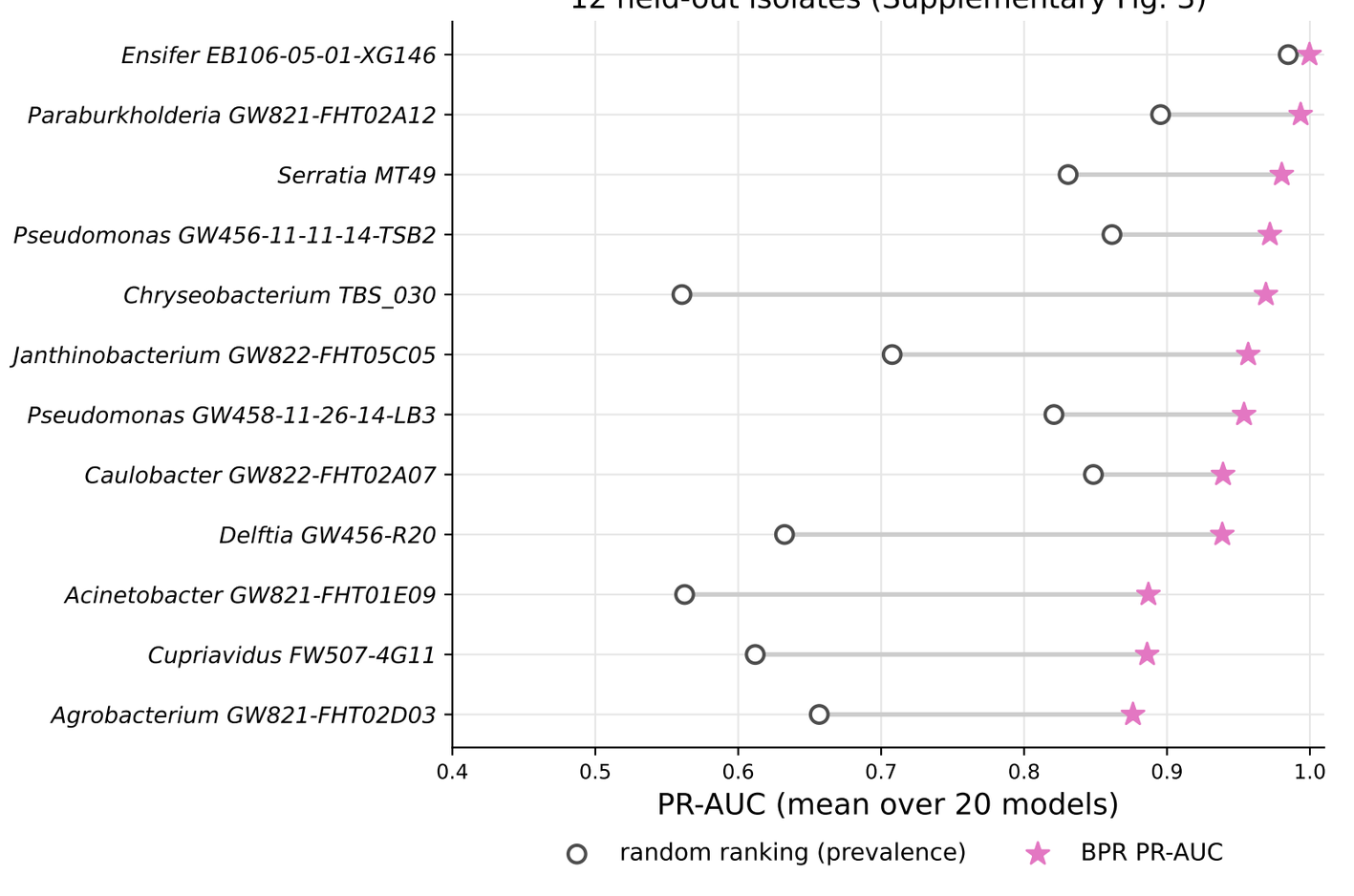
b


Fig. 1b hold-out, 32 dimensions, averaged over isolates (repeat 1)
Precision
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Recall
BPR (PR-AUC = 0.69)
WR-MF (PR-AUC = 0.64)
SVD (PR-AUC = 0.64)
LMF (PR-AUC = 0.60)
random ranking (0.29)

c


Fig. 3 hierarchical hold-out
PR-AUC on scoreable metabolites
1.0
0.8
0.6
0.4
0.2
0.0
Isolate
(n = 375)
Genus
(n = 213)
Family
(n = 325)
Order
(n = 320)
Hold-out level
BPR
naive baseline
random ranking (median prevalence)

d


12 held-out isolates (Supplementary Fig. 3)
Ensifer EB106-05-01-XG146
Paraburkholderia GW821-FHT02A12
Serratia MT49
Pseudomonas GW456-11-11-14-TSB2
Chryseobacterium TBS_030
Janthinobacterium GW822-FHT05C05
Pseudomonas GW458-11-26-14-LB3
Caulobacter GW822-FHT02A07
Delftia GW456-R20
Acinetobacter GW821-FHT01E09
Cupriavidus FW507-4G11
Agrobacterium GW821-FHT02D03
0.4
0.5
0.6
0.7
0.8
0.9
1.0
PR-AUC (mean over 20 models)
random ranking (prevalence)
BPR PR-AUC